



Data Structures and Program Design Using C

SCS 1301

AY 23 - Semester 1



Exercise III

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1. For the following exercises, use variants of **if-else** only:
 - (a) Write a program to check whether a given integer value is even.
 - (b) Extend the program in (a) above to check for odd integer values.
 - (c) Write a program to display if a given character is an upper or lower case letter.
 - (d) Extend the program in (c) above to recognize the case of the character and convert it to the opposite case. E.g., Upper case to lower case and vice versa.
 - (e) Extend the program in (d) above to output characters other than letters as “numbers” where it is a number and “Other Characters” for all other possibilities.
2. For the following exercises, use variants of **switch-case** only:
 - (a) Write a program to perform mathematical operations on two integer variables where the operator is given as a character variable. Consider addition, subtraction, multiplication, modulo, and division operators.
 - (b) Write a program that converts an integer from zero (0) to nine hundred and ninety-nine (999) given as a variable to words corresponding to the English pronunciation. Examples are given below.
 - 0 => zero
 - 98 => Ninety Eight
 - 711 => Seven Hundred and Eleven
3. For the following exercises, use **while** and **do-while** loops.
 - (a) Write a program to print all natural numbers from 1 to 50 using a while loop.
 - (b) Write a program to display the '\$' symbol 32 times using a while loop.
 - (c) Write a program to display all integers from 10 to 0 in reverse order using a while loop.
 - (d) Write a program to display all floating point numbers from 10.0 to -5.0 in 0.1 decrements.
 - (e) Write a program to display all counter numbers from 10 to 99. Indicate whether they are multiples of 2, 3, or 5. The output should match exactly as follows:

```
10      -multiple of 2      multiple of 5
11
12      -multiple of 2      multiple of 3
...
99      -multiple of 3
```

(f) Write a program to display all simple letters in the English alphabet.

4. For the following exercises, use the **for loop**.

(a) Write a program to display all integers from 20 to 35.

(b) Write a program to display the following pattern. Each loop should print one '*' per iteration.

```
*****
```

(c) Write programs to display the patterns below.

i. *

 *

 *

 *

 *

ii. *****

iii. *

 **

iv. *

 **

 **

 *

v. *

 *

vi. * 1
 ** 3
 *** 6
 **** 10
 ***** 15

 **
 *

vii. *****

5. What would be the output for each print statement in the code given below? You must dry run first and then execute the code to verify the results.

```
#include <stdio.h>
int main()
{
    double d;
    float f;
    long l;
    int i;

    i = l = f = d = 100/3;
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    d = f = l = i = 100/3;
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    i = l = f = d = 100/3;
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    d = f = l = i = (float)100/3;
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    i = l = f = d = (double)(100000/3);
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    d = f = l = i = 100000/3;
    printf("%d\t%l\t%f\t%f\n",i,l,f,d);

    return 0;
}
```

6. What would be the output for each print statement in the code given below? You must dry run first and then execute the code to verify the results.

```
#include <stdio.h>

int main()
{
    double d = 3.2, x;
    int i = 2, y;

    x = (y = d/i)*2;
    printf("%f\t%d\n", x, y);

    y = (x = d/i)*2;
    printf("%f\t%d\n", x, y);

    y = d * (x=2.5/d);
    printf("%f\n",y);

    x = d * (y = ((int)2.9+1.1)/d);
    printf("%f\t%d\n", x, y);
}
```

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